Evidence-Qualified Alignment of Pathology Foundation Models with Morphologic, Molecular, and Outcome Axes in Prostate Cancer

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Abstract

Pathology AI requires explanations that clinicians can verify, yet recoverable embedding signals do not necessarily drive predictions. We audited six non-interchangeable prostate-cancer axes in frozen CONCH and Virchow representations: Gleason/International Society of Urological Pathology (ISUP) grade, tumor phenotype/content, phosphatase and tensin homolog (PTEN) loss, androgen-receptor (AR) activity, speckle type BTB/POZ protein (SPOP) mutation, and recurrence. Grade and phenotype showed the strongest representation evidence and directional external transport. PTEN-related information was recoverable and AR activity showed positive pooled alignment, but their independent or site evidence remained unresolved. SPOP was unsupported, whereas recurrence changed with endpoint, representation, and clinical reference. Among the six axes, Grade/ISUP had the most complete evidence chain: removing an ISUP-correlated direction reduced locked biochemical-recurrence-head concordance for both encoders beyond matched random controls, while refitted-head intervals included zero. It was therefore the most developed functional example, not the sole organizing target. This comparative map provides an actionable audit vocabulary: transported coordinates can anchor clinician review of agreement and disagreement; conditional or unsupported axes indicate where explanations should be withheld; and evidence gaps prioritize external or functional validation. The study does not establish indispensable or externally transported functional use, clinical increment, or improved clinician outcomes.

1 Introduction

Clinical adoption of artificial intelligence requires calibrated reliance rather than blind trust or a persuasive explanation attached to every output. Clinicians should be able to judge when an AI output has a defensible basis, when it should be challenged, and where the model's competence ends. Explanations that fit the decision context can support rational checking [3], but familiar explanations can also increase reliance when AI advice is wrong [4]. A clinically recognizable explanation is therefore not evidence of trustworthy model behavior unless the proposed link can itself be tested.

Clinically established quantitative and ordinal targets offer a possible interface for such testing. A pathologist can define, measure, contest, and communicate grade or tumor extent; molecular laboratories can provide assay-defined targets; and clinical records provide outcomes. If an AI signal is demonstrably linked to one of these targets, the proposed basis becomes inspectable: a pathologist can confirm that the tissue supports the named feature, reject the explanation when it does not, or identify a meaningful discordance. Concept-based methods provide precedents. TCAV tests prediction sensitivity to human-defined concepts, and concept bottleneck models make named concepts explicit and intervenable [5, 6]. Medical-imaging and pathology systems have likewise exposed confounders or connected influential regions and embeddings to expert-recognizable findings [7, 8, 9]. These studies also show why concept labels, embedding similarity, and actual decision dependence cannot be treated as interchangeable.

Three evidence stages are particularly easy to collapse. A target may be recoverable from a frozen representation; that relationship may reproduce across cohorts and technical settings; and a downstream disease head may functionally use the target. Only the third directly tests sensitivity of a particular prediction rule, and even faithful functional use would not prove clinical benefit. We therefore order the evidence as recoverability, reproducibility and qualification, functional-use testing, and finally prospective evaluation of clinician reliance and utility. Each stage enables, but does not replace, the next.

Clinical orientation and reference availability. The disease studied here is predominantly prostate adenocarcinoma. Routine hematoxylin-and- eosin (H&E) sections expose gland architecture, tumor amount, and tissue context but do not directly measure DNA alterations or transcriptional activity. The six study axes therefore span different biological levels [10, 18, 19]. Grade/ISUP measures architectural differentiation; tumor phenotype/content measures tumor presence or amount; PTEN loss and SPOP mutation are binary molecular alterations; AR activity is a continuous transcriptional pathway score. Recurrence is a patient outcome observed after treatment rather than a microscopic structure graded on the slide. They are not six interchangeable measures of "cancer severity."

Crucially, the slides were model inputs, not the only study data. Every reported alignment estimate paired a slide-derived frozen embedding with a target-specific reference at the same analysis unit. The encoder was not fine-tuned on these study labels; a lightweight probe used the paired references to test whether the embedding contained recoverable target information. Reference availability nevertheless differed: morphology had compatible external labels, molecular references were largely confined to TCGA-PRAD, and recurrence definitions were non-equivalent across resources. Missing reference data were treated as not evaluable, not as evidence of non-alignment. Unsupported required an available comparison that did not qualify the proposed signal, as for SPOP. The detailed target, reference, availability, metric, and cohort frames are provided in Supplementary Section S1.

Why the six prostate-cancer axes are not interchangeable. A positive result supports a different claim at each level. Grade and phenotype are direct morphology targets; PTEN, SPOP, and AR are indirect histology–molecular associations; and recurrence is shaped by biology, treatment, follow-up, censoring, and endpoint definition. Recovering a target shows that related information is accessible in the embedding, not that H&E replaces a molecular assay or that a later prediction head uses the target. Conversely, failure to recover or transport a target warns against invoking it as an explanation, but does not prove absence of the underlying biology.

Existing studies often summarize image–label association with one pooled correlation or area under the receiver operating characteristic curve. Such a number cannot distinguish target information from grade composition, institutional case mix, acquisition conditions, sampling, or endpoint definitions [12, 13, 14]. Nor does it show whether a fitted relationship survives a new cohort without refitting. We therefore define *representation–target alignment* as patient-level association or discrimination between a registered target and the out-of-fold or locked external output of a simple probe fitted to a frozen CONCH or Virchow representation [1, 2]. We qualify each target across recoverability, external transport, conditional increment, site behavior, representation and sampling sensitivity, and endpoint fidelity, retaining these as an evidence vector rather than a composite score.

Position of this study. This study compares Grade/ISUP, tumor phenotype/content, PTEN loss, SPOP mutation, AR activity, and recurrence to determine which targets are accessible in frozen pathology representations and how their evidence changes across cohorts, clinical adjustment, sites, representations, sampling, and endpoint definitions. Grade and phenotype provide the broadest test of external transport; PTEN, AR, and SPOP distinguish recoverable, conditional, and unsupported molecular states; and recurrence tests whether outcome-linked information retains its meaning across event definitions and comparators. Among these axes, Grade/ISUP has the most complete evidence chain because it is additionally tested in two locked biochemical-recurrence heads as a deliberately narrow internal functional-sensitivity extension. ISUP is thus the most developed functional example, not the sole organizing target. The study does not explain a deployed model, establish indispensable or externally replicated functional use, comprehensively test clinical increment, or measure clinician reliance or benefit.

Its primary task is comparative: to map the evidential status of all six axes without treating their labels, biological levels, or available validation opportunities as equivalent.

Contributions. This paper makes four contributions. First, it defines clinically interpretable targets as *contestable shared coordinates* through which a pathologist can agree or disagree with a clinically stated basis for an AI output. Second, it provides a comparative six-axis map across cohorts and two foundation models, distinguishing transported morphology, context-sensitive molecular and outcome associations, and unsupported evidence. Third, it turns the map into an actionable audit vocabulary: transported coordinates can anchor clinician review of agreement and disagreement; conditional or unsupported axes indicate where explanations should be withheld; and evidence gaps prioritize external or functional validation. Fourth, it provides a worked extension from representation availability toward functional testing by applying targeted erasure and variance-matched controls to an ISUP-correlated direction in locked CONCH- and Virchow-derived risk heads. This final experiment extends rather than organizes the six-axis comparison. The proposed audit vocabulary is intended for model evaluation, communication, and validation planning; whether showing it to clinicians improves decisions remains untested.

2 Results

The six targets all concern prostate cancer, but they occupy different biological and clinical levels and therefore cannot be interpreted as interchangeable pathology features. They are Grade/ISUP, tumor phenotype/content, PTEN loss, AR activity, SPOP mutation, and recurrence. The primary result is their comparative evidence hierarchy; no single target serves as a surrogate for the other five. ISUP is highlighted later only because it reached an additional functional-sensitivity experiment. Every numerical alignment result used an available target-specific reference paired to the slide-derived embedding at the registered analysis unit. Unequal reference coverage limited which external, conditional, or functional questions could be asked; it was not converted into a negative association. Thus, not evaluated or blocked denotes a missing prerequisite, whereas unsupported denotes an available comparison that did not qualify the proposed signal. SPOP belongs to the latter category; phenotype functional testing and molecular external transport illustrate the former. Table 1 presents the study's headline evidence states and the permitted interpretation of each candidate coordinate. The Introduction provides the concise clinical orientation; Supplementary Section S1 retains the detailed target, paired-reference, availability, metric, and six-axis biological definitions and separates representation evidence from functional evidence about the BCR head.

Table 1: Evidence-qualified candidate shared coordinates. States summarize the available axis-specific evidence: transportable means directional external transfer was retained; context-sensitive means interpretation narrowed under a clinical, site, representation, sampling, or endpoint audit; unsupported means the frozen primary design did not qualify the signal. Functional use refers specifically to intervention on a locked BCR head and is not implied by recoverability.

Target	State	Permitted interpretation	Functional use
Grade/ISUP	Transportable	Grade information is recoverable and directionally transports across compatible external resources	Internal exploratory locked-BCR-head sensitivity; indispensable, external, and clinical-increment use not established
Tumor phenotype/content	Transportable	Morphology-linked phenotype is recoverable and transports directionally; the external target is grade-derived tumor/benign status, not exact tumor-content truth	Currently blocked: same-cohort independent tumor-content truth unavailable
PTEN	Context-sensitive	PTEN-related information is recoverable within TCGA-PRAD but grade-independent increment and external transport are not established	Feasible follow-up; not run
AR activity	Context-sensitive	AR activity has positive pooled alignment but grade-independent increment and site transport remain unresolved	Feasible follow-up; not run
SPOP	Unsupported (frozen design)	Evaluated against an available genomic reference, but no reliable frozen-design signal was established	Not qualified: recoverability unsupported
Recurrence	Context-sensitive	Recurrence alignment is endpoint representation and covariate hierarchy conditioned	Not applicable as input erasure; prognostic validation required

2.1 The study separates recoverability, reproducibility, and functional use

Alignment question. What evidence is required before a human-defined target can be used to interpret an AI representation, and which stronger conclusions remain outside that evidence?

Analysis frame. We organized each target by the evidence needed to use it as a human–AI interpretive coordinate (Figure 1). A target first had to be recoverable from a frozen representation at its registered analysis unit. External application without target-cohort refitting evaluated transport where compatible labels existed. Conditional, site, representation, and endpoint audits then identified restrictions. Unevaluated axes remained missing evidence rather than negative findings.

Evidence test. The test was not whether one metric exceeded a universal cutoff. Instead, the registered target was followed across the available evidence axes, with its cohort, analysis unit, metric, null, and uncertainty retained. External transport required an unchanged source probe; conditional increment required a held-out comparison with a clinical reference; site, setting, and endpoint audits addressed different alternative explanations.

Interpretive boundary. This structure produced three manuscript-facing states: transportable, context-sensitive, and unsupported in the frozen design. These are target-specific evidence states, not model confidence scores or clinical action thresholds. Recoverability and transport occupy the left side of the evidence chain. For grade, a separate internal FM6 analysis used locked BCR heads, target-specific erasure, and matched controls to test a narrow functional-sensitivity claim. The other targets have different functional-evidence states: phenotype is blocked by unavailable same-cohort independent truth; PTEN and AR are feasible follow-up analyses that were not run; SPOP lacks a qualified recoverable direction; and recurrence is the outcome predicted by the head rather than an input target for erasure. External functional replication was not achieved.

Qualification rule. A target was called transportable only when an appropriate external evaluation retained a consistent direction without target-cohort refitting. Context-sensitive means that the interpretation materially depended on grade, site, representation, sampling, or endpoint. Unsupported means that the frozen primary design did not support the proposed effect; it does not mean that the biology is absent. Missing axes remain not evaluated.

Figure 1: Evidence-qualified map of candidate shared interpretive coordinates. Rows are clinically interpretable targets and columns are distinct evidence axes. A supported cell means only that the named axis was supported. Unresolved, not evaluated, and not applicable cells are preserved. Cell codes are S, supported; C, context-sensitive; U, unresolved; NE, not evaluated; NA, not applicable; and X, unsupported in the frozen design. In the final column, IE denotes internal exploratory functional sensitivity for grade; BL, blocked because same-cohort independent phenotype truth was unavailable; NR, feasible but not run for PTEN and AR; NQ, not qualified because SPOP recoverability was unsupported; and NA, not applicable because recurrence is the predicted outcome. IE does not mean indispensable, externally reproduced, or clinically incremental use.

How to read the qualification map (Figure 1). Read each row from left to right rather than comparing colors vertically as if they were a model leaderboard. The first cells ask whether the target can be recovered and transported. The middle cells ask whether that interpretation narrows after clinical, site, technical, or endpoint checks. A supported cell applies only to the named axis; it does not fill a missing cell elsewhere. The final functional-use column is deliberately marked as not one generic "not tested" state: it distinguishes a completed internal exploratory test from an unavailable prerequisite, a feasible analysis not run, an unqualified direction, and a question that is not applicable. This prevents probe recoverability from being mistaken for functional evidence and prevents the scoped grade result from being generalized to other targets.

2.2 Morphology-linked targets provide the best-qualified candidate coordinates

Alignment question. Can clinically familiar architectural targets be recovered from frozen representations and retain their direction when the same probe is applied to independent resources?

Analysis frame. Grade and tumor phenotype were evaluated in NADT-Prostate and then transferred without target-cohort refitting to PANDA; PRECISE supplied a separate session-level grade estimate. Spearman correlation assessed rank preservation for grade and continuous phenotype, while AUROC assessed grade-derived benign-versus-tumor discrimination.

Evidence test. Grade and phenotype showed the broadest available alignment across cohorts (Figure 2). In NADT-Prostate, the patient-level CONCH grade probe had Spearman $\rho = 0.478$ (patient-bootstrap interval 0.170–0.712; 39 patients), and the tumor-phenotype signal had $\rho = 0.800$ (0.625–0.909; 39 patients). Applied without target-cohort refitting, the grade probe retained positive associations in PANDA Karolinska ($\rho = 0.354$; 565 case images) and Radboud ($\rho = 0.519$; 572 case images). The phenotype probe discriminated grade-derived tumor status in the same institutions with AUROCs of 0.786 and 0.871. A separate PRECISE evaluation yielded grade $\rho = 0.865$ across 17 imaging sessions.

Interpretive boundary. These rows demonstrate directional transport of morphology-linked information, not uniform precision. Saved target-cohort intervals were unavailable for PANDA and PRECISE and were not reconstructed. Analysis units also differed across patient, case image, and session. Exact effect sizes therefore remain specific to the registered cohort and unit. In particular, the PANDA phenotype target was grade-derived tumor-versus-benign status, not an independent same-patient measurement of quantitative tumor content. Its result supports directional transport of a tumor-related phenotype score; it does not show that exact tumor fraction was reproduced externally.

Qualification rule. Grade and phenotype qualify as the strongest transportable candidate coordinates because the locked source probe retained the expected direction in external resources. This state does not imply interchangeable effect sizes across datasets, complete representation explanation, or functional use by a later disease-prediction head.

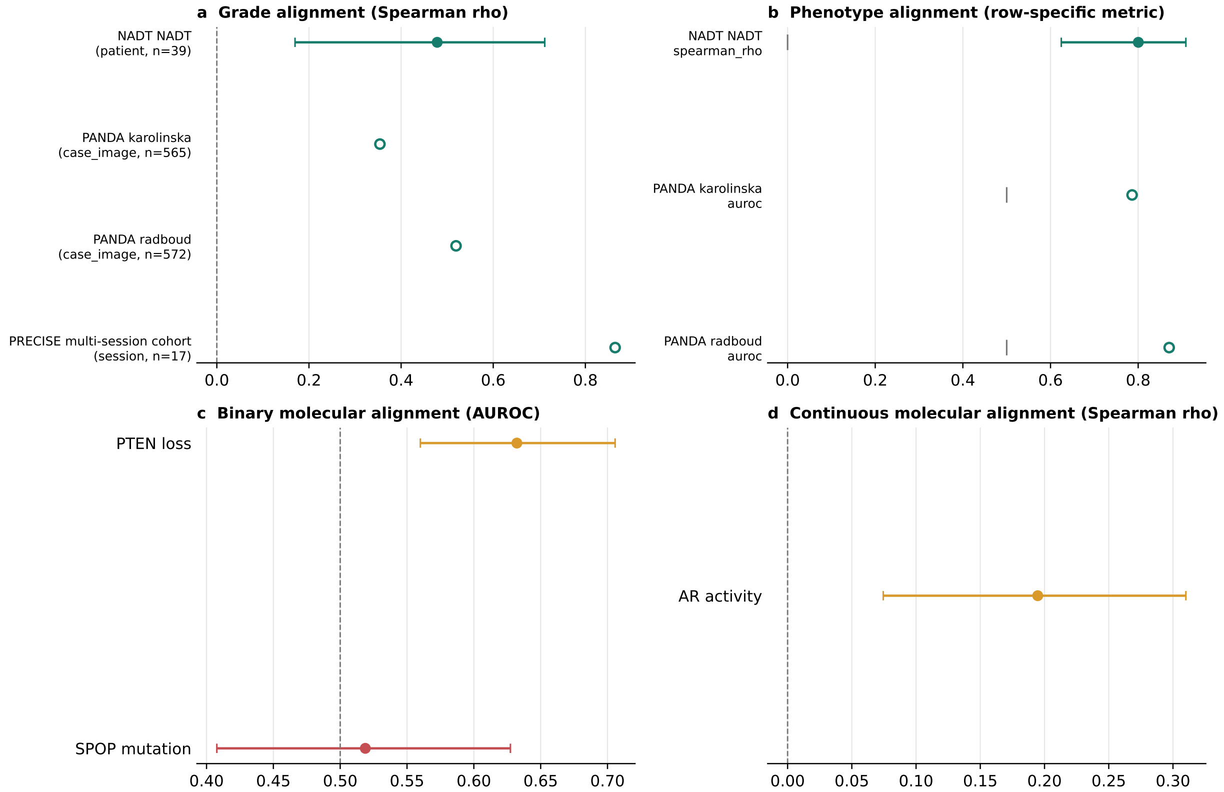

Figure 2: Recoverability and transport of known morphologic and molecular targets. **a**, Grade and phenotype estimates across source and external resources; metrics and analysis units are displayed rather than pooled. The PANDA phenotype rows use grade-derived tumor-versus-benign status and therefore test directional phenotype transport, not exact external tumor-content agreement. **b**, Frozen-primary molecular alignment in TCGA-PRAD; these molecular rows are within-cohort recoverability results, not external transport. Whiskers are shown only where saved patient-bootstrap intervals were available. Dashed lines mark correlation zero or AUROC chance.

The complete ten-row primary evidence frame, including cohort, unit, denominator, metric, and retained uncertainty, is provided in Supplementary Section S4. It prevents estimates with different analysis units or precision from being treated as directly interchangeable.

2.3 Molecular coordinates separate into recoverable, conditional, and unsupported states

Alignment question. Do frozen histology representations contain information associated with PTEN loss, SPOP mutation, and AR activity, and does any observed association provide information beyond grade or remain stable across source sites?

Analysis frame. The primary molecular frame comprised 273 TCGA-PRAD patients. PTEN loss and SPOP mutation were binary targets scored by AUROC; AR activity was continuous and scored by Spearman correlation. Nested, patient-disjoint models compared image information with grade, while site and representation audits tested the scope of the pooled association.

Evidence test. PTEN loss was recoverable with a frozen-primary CONCH AUROC of 0.632 (0.560–0.706). AR activity had a positive patient-level Spearman correlation of 0.195 (0.074–0.310). By contrast, the SPOP frozen-primary AUROC was 0.519 (0.408–0.627), and its broader correlated seed-cell range crossed chance (0.348–0.679). Conditional audits further narrowed PTEN and AR (Figure 3). The held-out increment beyond grade was +0.019 AUROC (−0.025 to

+0.068) for CONCH PTEN and +0.035 (−0.013 to +0.087) for Virchow PTEN. AR increments in R^2 were +0.004 (−0.036 to +0.042) and +0.019 (−0.015 to +0.051), respectively. The pooled AR direction was positive, but each of six stored site-specific intervals crossed zero. The expanded site audit in Figure 3 also preserves all evaluable SPOP sites: their intervals included or touched chance, while the CH site contained no positive event and therefore had undefined AUROC. Site remained entangled with acquisition and case mix.

Interpretive boundary. Recoverable molecular information was not shown to add held-out information beyond grade. Site codes cannot be given a causal scanner interpretation. None of these results establishes routine molecular diagnosis from H&E or functional use of a molecular target by an AI disease decision. The SPOP genomic reference was available and was used; its unsupported state is therefore an evaluated lack of qualification in this frozen design, not a missing- label or not-evaluated result.

Qualification rule. PTEN and AR are context-sensitive candidate within-cohort coordinates because primary recoverability was present but independent increment beyond grade and site transport were not established. SPOP is unsupported in the frozen primary design because its estimate and broader setting range did not separate reliably from chance. These states qualify the present evidence, not the biological existence of the alterations.

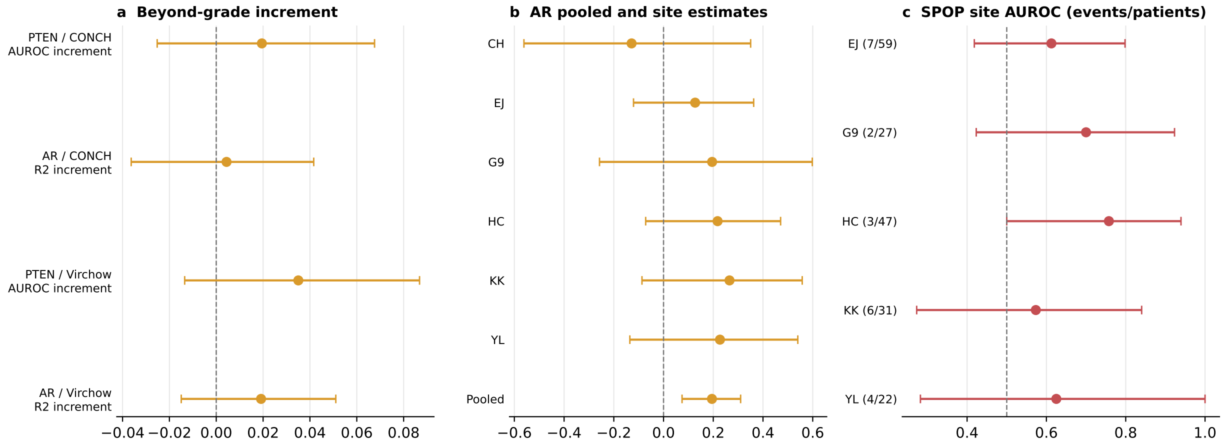

Figure 3: Conditional and site-specific narrowing of molecular alignment. **a**, Held-out PTEN and AR increments beyond grade for CONCH and Virchow. Positive values indicate improvement after adding the image-derived score to the grade reference; the row-specific metrics are AUROC for PTEN and R^2 for AR. **b**, Pooled and site-specific AR correlations. **c**, SPOP site-specific AUROCs, with positive-event and patient counts; CH is retained as a single-class, undefined row in the accompanying source table rather than assigned an AUROC. Intervals crossing or touching the relevant null indicate unresolved conditional evidence, not absence of biological signal. PTEN and AR use by a BCR head was not tested. SPOP was tested against an available genomic reference but did not qualify a stable frozen-representation direction.

2.4 Outcome alignment changes with endpoint and clinical reference

Alignment question. Does a transferred image-derived recurrence score preserve patient risk ordering, and is its interpretation stable when the event definition or clinical reference changes?

Analysis frame. The post-hoc score was fitted in LEOPARD and applied unchanged to TCGA-PRAD. Concordance was evaluated separately for reconstructed with-tumor recurrence and official TCGA-CDR PFI. Increment analyses compared grade alone with grade plus the image score in the same 153 complete cases and the same paired bootstrap draws.

Evidence test. In the 270-patient transfer frame, the reconstructed with-tumor endpoint yielded a C-index of 0.673 (95% interval 0.587–0.759; 57 events), whereas official PFI yielded 0.586 (0.482–0.689; 42 events). On 153 complete cases, adding the image score to grade changed C-index by +0.083 (+0.012 to +0.157; 30 reconstructed events), but the paired integrated Brier-score interval included zero. For official PFI, the C-index increment was +0.032 (−0.040 to +0.126; 15 events), and all paired official-PFI C-index and integrated-Brier-score intervals included zero. Fuller clinical and site comparisons did not establish robust independent increment. The complete same-patient, same-bootstrap-draw comparison family is reported in Supplementary Section S9; it shows that interpretation depends on the comparator and performance metric. Figure 4 shows that complete comparison family while retaining separate endpoint panels.

Interpretive boundary. Concordance measures risk ordering, not calibrated individual risk or clinical usefulness. The reconstructed endpoint and official PFI contain different event definitions and event counts; performance on one cannot be relabelled as performance on the other. The PCA sweep that generated the post-hoc signal was exploratory rather than an unbiased model-selection procedure.

Qualification rule. Recurrence remains an endpoint-conditioned, context-sensitive exploratory coordinate because its estimate changed with the endpoint and robust independent increment over clinical references was not established. It is not a validated prognostic biomarker.

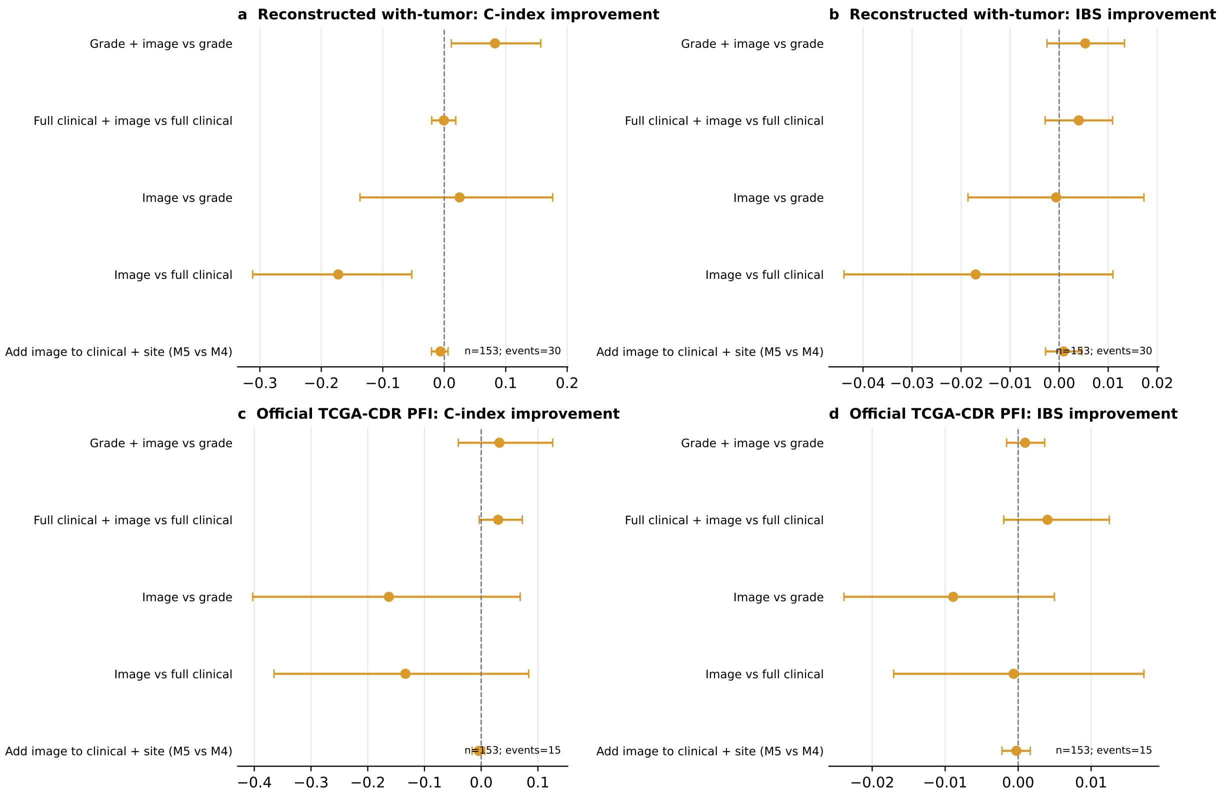

Figure 4: Complete paired outcome-comparison family across non-equivalent endpoints. Rows compare the image score with grade or fuller clinical references, or compare nested clinical models, on the same patients and bootstrap draws. **a,b**, Reconstructed with-tumor C-index and integrated-Brier-score (IBS) improvements. **c,d**, The same comparisons for official TCGA-CDR PFI. Positive values favor the first, augmented, or more complete model under the registered delta definition. These panels are comparator and endpoint audits, not independent model validations; the difference between reconstructed recurrence and official PFI is part of the result rather than a nuisance to be pooled.

2.5 Representation audits bound sensitivity without selecting a universal model

Alignment question. Would the target-level interpretation change if the frozen encoder, physical image scale, number of sampled tiles, or random tile sample changed?

Analysis frame. Six targets were evaluated across 12 encoder–scale–tile configurations and five sampling seeds, yielding 360 correlated seed cells. The audit also contained 390 seed-matched scale, encoder, and tile-budget contrasts.

Evidence test. For every target and configuration, the five seed-specific scores were compared with the appropriate null—zero for Spearman correlation and 0.5 for AUROC or C-index. Paired contrasts preserved the same seed when comparing encoder, scale, or tile budget. Grade, phenotype, PTEN, and AR retained their direction across the observed configuration ranges. SPOP ranges crossed the null in 8 of 12 configurations and recurrence in 1 of 12.

Interpretive boundary. These repeats reused cohorts, patients, folds, and representations. They reveal whether an interpretation is dependent on analytic settings, but they are not independent replication and do not support a universal CONCH-versus-Virchow ranking. Complete configuration ranges, null crossings, and all 390 paired contrasts are reported in Supplementary Section S8; Figures 5 and 6 expose their target-specific ranges and contrast directions in the Main manuscript.

Qualification rule. Setting variation narrows a target's interpretation when its observed range crosses the metric null or paired contrasts show material dependence. Stable direction across this grid supports technical robustness within reused data only; external reproducibility still requires a different cohort.

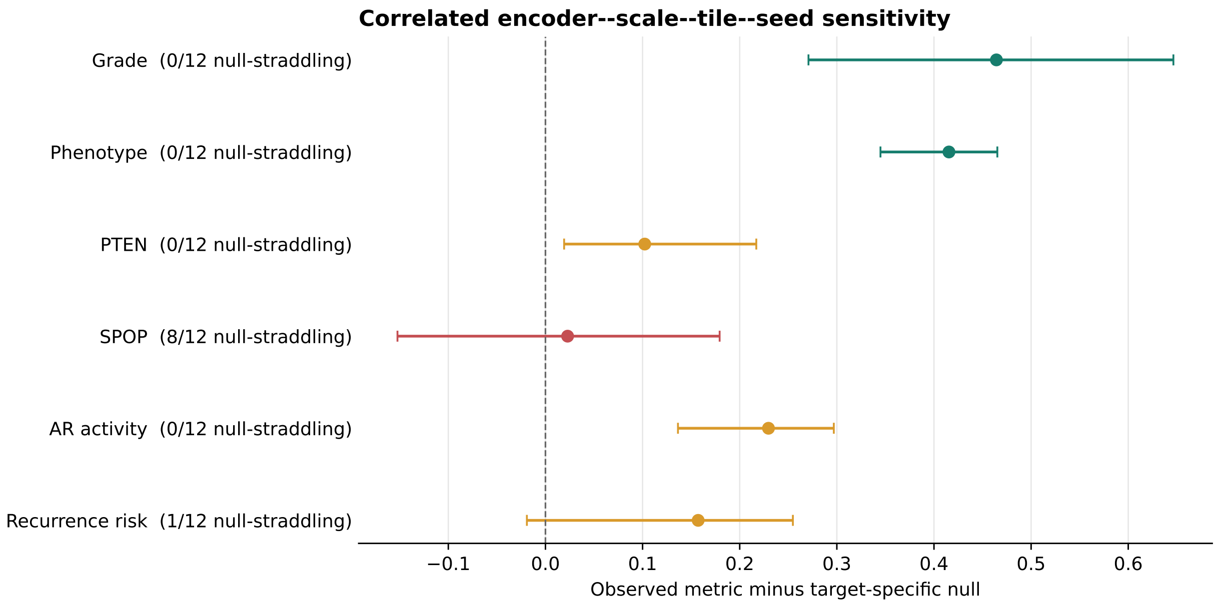

Figure 5: Target-specific representation and sampling sensitivity across all six axes. Observed five-seed ranges across 12 encoder–scale–tile configurations are shown relative to each target's metric null. Counts record configurations whose observed seed range straddled the null: 8 of 12 for SPOP and 1 of 12 for recurrence, versus retained direction across the observed ranges for grade, phenotype, PTEN, and AR. All six axes are displayed on their own metric scales; the 360 cells are correlated internal sensitivity settings rather than independent validations or external transport tests.

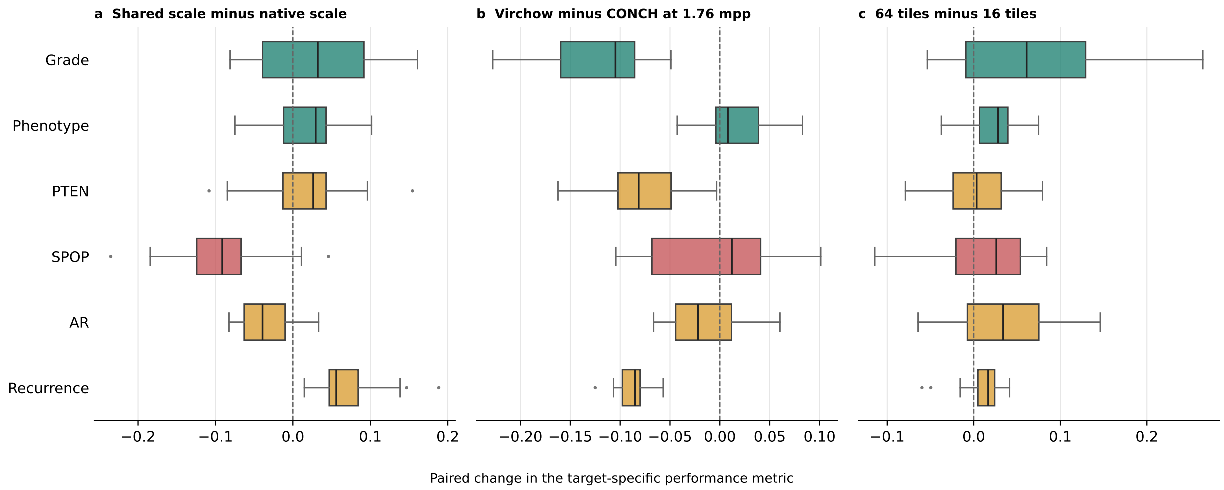

Figure 6: Seed-matched encoder, physical-scale, and tile-budget contrasts. Each box summarizes paired performance changes for one target while preserving the same sampling seed. **a**, Shared physical scale minus the encoder-native scale. **b**, Virchow minus CONCH at the shared 1.76-micrometre-per-pixel scale. **c**, 64 tiles minus 16 tiles per slide. The vertical null denotes no setting effect; direction is specific to the named comparison and does not define a universally preferable encoder or setting. All 390 comparisons reuse cohorts and folds.

2.6 ISUP provides the most developed extension from representation alignment to internal functional sensitivity

The preceding analyses compared the representation-level evidence for all six axes. We next asked how far one sufficiently recoverable axis could be advanced toward functional-use testing. Grade/ISUP was the only target for which the current data and protocol supported this extension; it was not treated as the sole or a priori organizing target of the study.

Alignment question. Does a locked AI risk head change when an ISUP-correlated direction is removed from its input, and is that change larger than removing an equally large but randomly oriented part of the representation?

Analysis frame. The exploratory FM6 frame comprised 392 TCGA-PRAD patients with 80 BCR events. CONCH and Virchow remained frozen. For each encoder, a simple ridge probe learned an ISUP-correlated direction, while a ridge-penalized Cox head learned BCR risk from the frozen representation. "Locked head" means that after this risk rule had been fitted, its coefficients were held fixed while the ISUP-correlated direction was removed; the foundation model itself was not retrained. The same patient-disjoint folds were used throughout.

Evidence test. ISUP was recoverable out of fold from both representations (Spearman ρ = 0.615 for CONCH and 0.658 for Virchow). The full BCR heads had C-indices of 0.627 (0.553–0.694) and 0.632 (0.564–0.691), respectively. Removing the ISUP-correlated direction while keeping the head fixed reduced concordance by 0.041 (0.011–0.073) for CONCH and 0.023 (0.007–0.040) for Virchow. Both changes exceeded the respective variance-matched random-direction controls (p = 0.0099 for each encoder). Thus the observation repeated across two distinct frozen representations under the same internal analysis contract.

A protocol-locked clean rerun regenerated the FM6 analysis and figure outputs without changing the cohort, embeddings, folds, seeds, endpoints, controls, thresholds, or hyperparameters. All 20 protocol-defined nonvolatile output SHA-256 hashes matched the pre-run baseline exactly, and the estimates, intervals, and gate decisions above were unchanged. This establishes computational reproducibility of the internal experiment; it is not an independent-cohort replication.

Interpretive boundary. The result supports sensitivity of these two internal heads to an ISUP-correlated subspace; it does not prove that either head implements a human grading rule. When each head was refitted after erasure, the full-minus-erased differences narrowed to 0.014 (−0.006–0.035) for CONCH and 0.010 (−0.003–0.022) for Virchow, with intervals including zero. This indicates that other embedding directions may compensate, so indispensable use was not established. A limited comparison of ISUP plus the AI score against ISUP alone also had intervals including zero (CONCH, −0.000, −0.039–0.039; Virchow, −0.032, −0.088–0.025). That boundary check is insufficient to conclude that clinical increment is absent; a dedicated, externally validated clinical-increment study was not performed.

The analysis used whole-tissue representations. A remediated tumor detector reached specificity 0.810 on the already opened SICAP secondary test, which we accept as internal technical evidence. However, sensitivity was only 0.562 and 0.679 on newly held-out PANDA Karolinska and Radboud providers. Independent-domain sensitivity therefore remains unresolved, and the present result is not described as tumor-specific or externally transported functional use.

Qualification rule. Grade receives an *internal exploratory functional-sensitivity* state because the fixed-head effect was direction-specific, exceeded matched controls, and repeated across CONCH and Virchow. It is the most developed example in the comparative map, but it is not promoted to indispensable, external, mechanistic, or clinically incremental use. No residual or unknown representation direction is evaluated as a new marker in this manuscript. In the locked FM6 gate vocabulary, both encoders passed internal whole-tissue recoverability (R), disease association (A), and exploratory functional sensitivity (U); external transport (T) was not tested, and a strong H2 functional-use claim remained prohibited.

2.7 The final alignment map distinguishes best-qualified, conditional, and unsuitable coordinates

Alignment question. After combining all available axes, which targets are best-qualified candidates for shared human–AI coordinates, and what evidence is still required?

Analysis frame and evidence test. We synthesized the registered primary, external, conditional, site, representation, and endpoint rows without averaging unlike metrics or converting missing evidence into failure.

Interpretive boundary and qualification rule. The combined evidence supports a practical hierarchy. Grade and phenotype are the best-qualified candidate shared coordinates because they were recoverable and directionally transported. PTEN and AR can orient interpretation within the evaluated cohort, but their independence and transport remain unresolved. SPOP is unsuitable under the frozen primary design, and recurrence remains exploratory because its meaning changes with endpoint and reference model. Grade alone also has internal exploratory functional-sensitivity evidence; phenotype and the other targets do not. Table 1 summarizes these decisions; expanded biological definitions are retained in Supplementary Section S1.

Figure 7 translates this hierarchy into a three-layer linkage map: the clinically interpretable coordinate, the operational AI feature through which it was tested, and the available evidence about downstream judgment. Here, an "AI feature" means a probe-defined direction or score in the frozen embedding, not a localized gland, cell, or latent unit with an independently established biological meaning. Grade/ISUP is the only pathology coordinate that reaches the third layer through a locked-head intervention. The other dashed links end at representation evidence, but for different reasons: phenotype is currently blocked by the absence of same-cohort independent tumor-content truth; PTEN and AR are feasible follow-up tests that were not run; SPOP lacks a qualified representation feature; and recurrence is the prediction outcome rather than an input-feature erasure target. This does not mean that only ISUP describes prostate cancer or that all five remaining signals were equally visible. Phenotype had strong transported representation evidence; PTEN and AR had conditional representation evidence; SPOP did not yield a qualified representation feature; and recurrence produced an endpoint-sensitive risk association. What is unique to ISUP is evidence that the tested BCR heads were functionally sensitive to its correlated embedding direction.

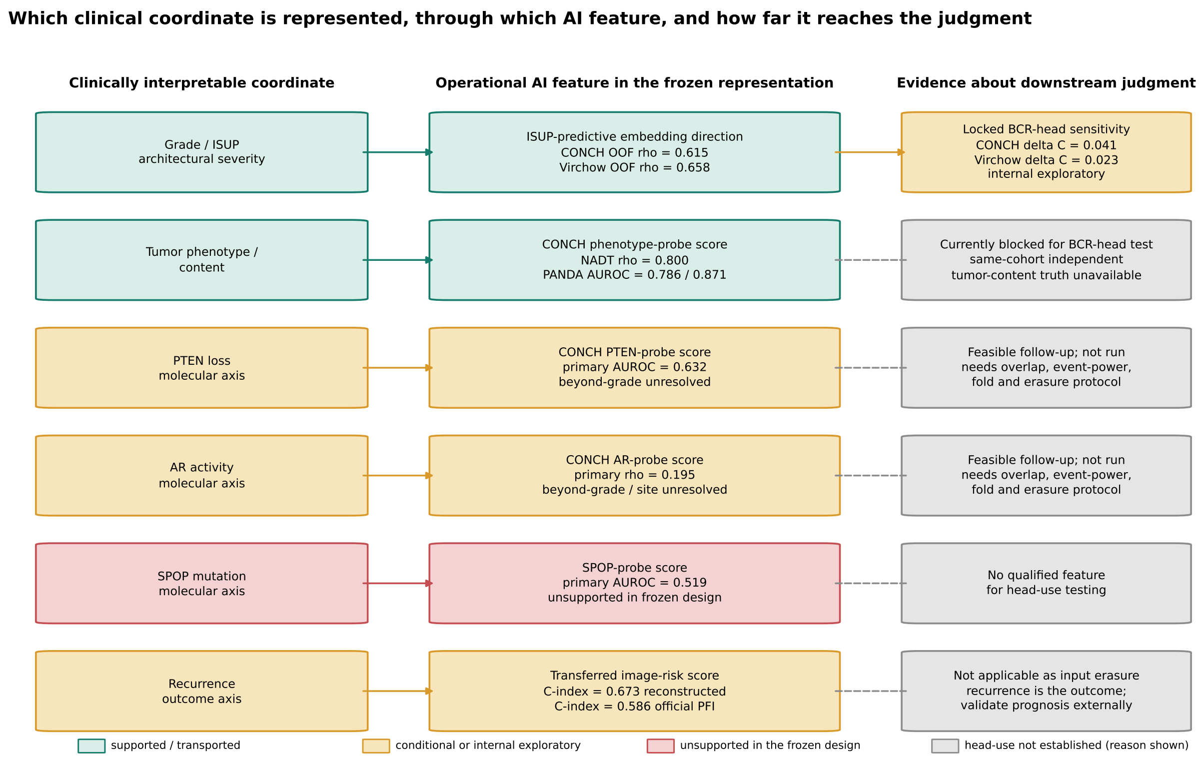

Figure 7: Human quantitative coordinate–AI feature–judgment linkage map. The left column names the clinically interpretable morphologic, molecular, or outcome axis. The middle column identifies the operational representation feature used in this study: a target-predictive probe direction or image-derived score. The right column shows whether that feature was linked to a downstream judgment. Solid arrows denote an evaluated link; dashed gray links denote that head use was not established, with the row-specific reason written in the right-hand box. Only ISUP has internal locked-BCR-head erasure evidence, repeated in CONCH and Virchow. This figure does not claim that only ISUP characterizes cancer, that every other target is absent from the representation, or that a probe direction is a localized histologic primitive. Phenotype is strongly represented, PTEN and AR are conditionally represented, SPOP is unsupported, and recurrence is an endpoint-sensitive outcome signal; these states must not be collapsed into one negative category. The displayed coordinates do not completely explain an AI decision.

3 Discussion

The central result is not that one prostate-cancer target explains a foundation model. It is that six clinically interpretable axes occupy different, empirically distinguishable positions in a shared-coordinate map. Grade/ISUP and tumor phenotype/content provided the broadest representation-level transport; PTEN loss and AR activity were detectable but became conditional under clinical or site audits; SPOP mutation did not yield a qualified direction; and recurrence changed with endpoint, comparator, encoder, and scale. These contrasts are the main scientific finding. The ISUP intervention is important because it extends the best-developed axis one step further, from representation availability to internal functional sensitivity, but it does not make ISUP the sole organizing target.

This hierarchy must be read together with reference availability. The study did not ask a slide-only model to agree with six unobserved quantities: each reported estimate compared a slide-derived embedding with an available, target-specific paired reference. What differed was the breadth of those references. Grade and phenotype had the widest compatible external coverage; PTEN, SPOP, and AR molecular references were largely confined to TCGA-PRAD; and recurrence references used non-equivalent event definitions. Missing external or same-patient functional truth therefore produced not-evaluated or blocked cells, not evidence of non-alignment. By contrast, SPOP was unsupported after comparison with an available genomic reference. The observed hierarchy consequently reflects both biological accessibility in H&E and unequal opportunities for transport and functional validation.

The morphology results illustrate both the promise and the incompleteness of a shared coordinate. Grade describes architectural differentiation, whereas phenotype/content describes the presence or amount of tumor-like tissue. Both were recoverable and retained direction under compatible external application without target-cohort refitting, making them the strongest current candidates for cross-cohort interpretation. Their different clinical meanings remain essential: a model can encode tumor amount without encoding aggressiveness, and can encode grade without measuring tumor burden. Transport also does not show that either coordinate completely explains the representation or is used by every downstream task. Moreover, PANDA phenotype transport used grade-derived tumor-versus-benign status rather than an independent quantitative tumor-content assay, so it supports a transported tumor-related direction but not externally reproduced exact tumor fraction.

The molecular targets show why a positive pooled estimate is not a sufficient explanation. PTEN-related information was recoverable in the frozen primary analysis (AUROC 0.632), and AR activity showed positive pooled alignment (Spearman $\rho = 0.195$). Yet held-out increments beyond grade included the null, and the AR site intervals did not establish site transport. These findings support conditional histology–molecular associations, not routine molecular diagnosis from H&E and not evidence that a BCR head used PTEN or AR. SPOP provides the complementary negative qualification: its primary estimate and setting range did not support a stable recoverable direction. That result limits SPOP-based interpretation in this design without implying that SPOP biology is absent or unimportant.

Recurrence occupies a different level again. It is a post-treatment patient outcome shaped by disease biology, treatment, follow-up, censoring, and the event definition rather than a microscopic feature directly assigned to the slide. The transferred image score ranked risk more strongly for reconstructed recurrence than for official PFI, and its apparent increment changed with the clinical reference and metric. Recurrence is therefore evidence that an embedding can carry endpoint-associated risk information, but it is not an interchangeable sixth pathology feature or a validated prognostic biomarker. Keeping the endpoints separate prevents a favorable result under one definition from being silently generalized to another.

The evidence-qualified design explains this hierarchy more faithfully than one pooled accuracy number. Recoverability asks whether target-related information is accessible; external transport asks whether the same fitted rule retains direction elsewhere; conditional and site audits ask whether clinical structure or acquisition context narrows the interpretation; and representation, sampling, and endpoint audits test different forms of dependence. CONCH and Virchow are therefore complementary probes of target-specific representation sensitivity rather than contestants in a universal model ranking. Likewise, the 360 seed cells and 390 matched setting contrasts expose analytic sensitivity but do not create new patients, institutions, or independent replications.

This position differs from methods that directly connect concepts to a trained prediction. TCAV estimates prediction sensitivity along a concept direction, and concept bottleneck models route predictions through explicit, intervenable concepts [5, 6]. Recent medical-imaging and pathology systems have similarly linked model-important regions or concept embeddings to expert-recognizable findings [8, 9]. Here, the primary contribution is comparative qualification of pre-existing clinical targets in frozen representations. The targeted-erasure experiment is a secondary worked example of how a qualified coordinate may be tested against a downstream decision; it is not a complete concept-grounded architecture or a mechanistic-fidelity result.

Among the six axes, Grade/ISUP had the most complete available evidence chain. Removing an ISUP-correlated direction reduced fixed-head concordance more than variance-matched random removal for both CONCH and Virchow, arguing against an idiosyncrasy of one representation. Refitted heads could nevertheless compensate within statistical uncertainty, and the limited ISUP-only comparison was not designed or powered as a comprehensive clinical-increment study. The warranted claim is therefore internal sensitivity of two BCR heads to an ISUP-correlated subspace, not necessary use of a human-equivalent grading mechanism, external functional transport, or absence of clinical increment.

Exact agreement of all 20 protocol-defined nonvolatile output hashes in the locked clean rerun strengthens the computational reproducibility of this internal result. It does not add new patients, institutions, outcomes, or labels and therefore cannot upgrade the claim to external functional replication.

The other five axes stop at different points for different reasons. Phenotype/content has strong transported representation evidence, but its BCR-head intervention is blocked by the absence of independently validated same-patient tumor-content truth in the TCGA BCR frame and by unresolved independent-domain detector sensitivity. PTEN and AR functional-use tests are feasible in principle but were not run; they require a newly locked molecular-label–BCR intersection, adequate event counts, folds, directions, matched controls, and preferably an external molecular–BCR cohort. SPOP was not advanced because a reliable intervention direction was not qualified. Recurrence is the outcome predicted by the BCR head, so its appropriate next test is external prognostic validation rather than erasure as though it were an input pathology coordinate. Absence of a functional-use result is therefore not one common negative result, and it does not show that a head ignores PTEN, AR, phenotype, or any other untested axis.

Several limitations bound the comparative hierarchy. External target-matched cohorts were available mainly for morphology, so stronger transport there partly reflects greater opportunity for testing. Some external rows lacked retained interval estimates, several molecular binary rows lacked preserved event-count accounting, and site remained confounded with acquisition and case mix. The recurrence signal was post hoc, endpoint definitions differed, and paired analyses used complete cases with modest event counts. The remediated tumor detector met the accepted secondary SICAP specificity threshold, but sensitivity remained below the prespecified threshold in both held-out PANDA providers; the ISUP functional result must therefore remain whole-tissue and internal. None of the six axes was evaluated for calibration, treatment utility, autonomous diagnosis, or population-wide generalization. Clinician trust, reliance, diagnostic accuracy, and the effect of presenting these coordinates were not measured; any benefit for calibrated reliance remains a prospective hypothesis. Clinically familiar explanations can themselves increase inappropriate reliance [4], which makes that prospective test necessary.

Joint completeness and residual-marker discovery are deliberately separate next questions. Completeness would require two or more independently measured clinical or pathology targets on the same patients, a pre-specified joint intervention, and adequate external outcome data. A new residual marker would additionally require removal of known targets and technical confounders, repetition across models and independent cohorts, an explicit measurable marker definition, and biological or clinical validation. Separating those studies preserves the present contribution: a comparative map of what is currently shared, what is conditional, what is unsupported, and which evidence step each target must satisfy next.

4 Conclusion

Clinically interpretable morphologic, molecular, and outcome axes can provide a common language for examining pathology-foundation-model representations, but they do not all provide the same kind or strength of evidence. Across six targets, Grade/ISUP and tumor phenotype/content formed the strongest transported morphology-linked coordinates; PTEN loss and AR activity provided conditional within-cohort molecular coordinates; SPOP mutation was unsupported in the frozen primary design; and recurrence remained conditioned by endpoint, clinical reference, representation, and scale. This comparative hierarchy—rather than the performance of any single target—is the primary conclusion.

Phenotype transport should be read as directional agreement with a grade-derived external tumor-versus-benign reference, not as validation of exact quantitative tumor content. SPOP, by contrast, was evaluated against an available genomic reference and remained unsupported; it was not simply unmeasured. These two cases illustrate why missing evidence and evaluated non-qualification must remain separate.

The hierarchy does not treat unavailable labels as negative results. Every reported alignment estimate used a paired target reference, but external and same-patient functional references were unevenly available. Accordingly, not evaluated or blocked identifies a missing validation prerequisite, whereas unsupported identifies an available comparison that did not qualify the signal. This distinction is necessary before the map can be used to audit an AI explanation.

The experiments support four connected contributions. First, they show that familiar clinical targets can be treated as contestable coordinates whose presence, absence, and limits can be tested in frozen representations. Second, the contrasting six-axis results show why recoverability, external transport, clinical independence, site behavior, representation sensitivity, and endpoint fidelity must be kept separate. Third, the qualified map turns those evidence differences into an actionable audit vocabulary. Fourth, Grade/ISUP provides the most developed example of the next evidence step: targeted removal of an ISUP-correlated direction reduced fixed BCR-head concordance for both CONCH and Virchow more than matched random removal. Refit intervals including zero show that the direction was influential but not established as indispensable. ISUP therefore illustrates how a qualified coordinate can be advanced toward functional testing; it does not define the study's only relevant coordinate or prove a human-equivalent decision mechanism. The locked clean rerun's 20/20 nonvolatile hash agreement establishes exact computational reproducibility of that internal analysis, not external functional replication.

The map therefore has a practical use before clinical deployment. Transported coordinates can anchor clinician review of agreement and disagreement between an AI-derived signal and the recorded tissue or assay target. Conditional or unsupported axes indicate where explanations should be withheld rather than presented as familiar but unverified reasons for a prediction. Evidence gaps prioritize external or functional validation, making explicit which target, cohort, intervention, or endpoint must be tested next. These are uses for model audit, communication, and study design; they are not evidence that displaying the map already improves clinician decisions.

At the clinical-use level, pathologist agreement, calibrated reliance, diagnostic benefit, and patient outcomes were not evaluated. Functional-use evidence was also not uniformly available: phenotype was blocked by missing same-cohort independent truth, PTEN and AR were feasible follow-ups not run, SPOP lacked a qualified intervention direction, and recurrence requires prognostic validation rather than input erasure. The paper therefore demonstrates a partial, auditable human–AI interface, not completed clinical validation.

The central claim is supported within that boundary: clinically interpretable targets were not shown to completely explain AI representations or decisions, but a comparative and explicitly qualified map identifies supported, conditional, and unsupported overlap. Grade and phenotype provide the clearest representation examples, while ISUP supplies one scoped internal functional-sensitivity example. This map does not itself demonstrate a new residual marker; it defines a discovery space. A residual should advance only if, after jointly accounting for known clinical–pathology targets and technical confounders, it recurs across models and independent cohorts, contributes to a locked prediction head, and becomes a reproducible human-measurable feature with biological or clinical validation. The present study qualifies shared coordinates; external functional validation, joint completeness, clinician studies, and residual-marker discovery remain distinct follow-up studies.

5 Methods

The Methods retain the information needed to interpret and reproduce the main claims; implementation details, complete sensitivity families, and missingness accounting are provided in Supplementary Sections S1–S12. The analyses, endpoints, folds, and frozen estimates were inherited from the source prostate-biomarker study without retrospective optimization for this manuscript.

5.1 Study design and interpretation contract

Representation–target alignment was defined as association or discrimination between a registered target and the patient-disjoint out-of-fold or locked external output of a probe fitted to a frozen representation. The primary design compared Grade/ISUP, tumor phenotype/content, PTEN loss, SPOP mutation, AR activity, and recurrence under target-appropriate metrics and qualification axes. It tests whether target-related information is recoverable and reproducible, not whether a disease-prediction head uses the same human concept. A secondary extension tested Grade/ISUP with locked BCR heads, targeted erasure, and matched controls because this axis had the required recoverable direction and available internal outcome frame. This unequal testing depth is reported as part of the six-axis evidence map rather than interpreted as evidence that grade is uniquely important. Neither analysis establishes mechanistic equivalence, indispensable use, complete explanation, clinical utility, or an observed effect on clinician trust or reliance. Clinician trust, reliance, diagnostic performance, and clinical utility were not study endpoints.

Targets were evaluated on the evidence axes permitted by their data: primary recoverability, cross-cohort transport, conditional increment, site behavior, representation and sampling sensitivity, and endpoint fidelity. The resulting states—transportable, context-sensitive, unsupported in the frozen design, unresolved, not evaluated, or not applicable—are evidence summaries rather than model-confidence or clinical-action scores.

5.2 Cohorts, targets, and analysis units

NADT-Prostate supplied source grade and tumor-phenotype labels; PANDA supplied Karolinska and Radboud case images and official ISUP grades for locked transport; PRECISE supplied session-level pathologist Gleason scores; TCGA-PRAD supplied histology, grade, PTEN loss, SPOP mutation, AR activity, and recurrence-style endpoints; and LEOPARD supplied BCR labels for fitting the transferred recurrence score [11, 10, 17, 15, 16]. Only evaluable target rows were used, and missing or uncertain outcomes were not recoded as negative. Patient, case-image, session, and slide effects remained explicitly distinct.

For NADT, predictions and labels were aggregated by patient and phenotype was the mean tumor fraction. PANDA sampling capped each institution–ISUP stratum at 100 cases before tissue filtering; its phenotype target was ISUP 0 versus ISUP at least 1. Recurrence-style definitions were not pooled or renamed: reconstructed with-tumor recurrence, strict recurrence, TCGA-CDR PFS and DFS, and official PFI retained separate identities [19]. Full cohort, target, event, and analysis-unit definitions are given in Supplementary Sections S1 and S9.

5.3 Frozen representations, probes, and evaluation

CONCH and Virchow encoder weights remained frozen. Tissue-qualified tiles were converted to embedding vectors and averaged into slide representations; patient endpoints then used patient aggregation. The sensitivity program crossed encoder, native or shared physical scale, tile budgets of 16, 32, or 64, and five sampling seeds on fixed cohorts and folds.

Continuous targets used standardized ridge regression and binary TCGA-PRAD targets used standardized logistic regression. Ridge regression is a linear model whose coefficient penalty stabilizes fits when thousands of correlated embedding features are available for far fewer patients; the fitted coefficients define a target-predictive direction rather than a localized gland or cell type. Logistic regression converts a weighted embedding score to a binary probability. Within-cohort evaluation used five patient-disjoint grouped folds, so every patient was scored by a probe fitted without that patient's data. Locked transfer applied the source probe and preprocessing to the target cohort without refitting.

Metrics followed the clinical target type. Continuous or ordinal targets used patient Spearman correlation, which asks whether predicted and observed ranks move together. Binary targets used AUROC, a threshold-free ranking measure rather than classification accuracy at one cutoff; AUROC does not choose a clinical sensitivity/specificity threshold. Survival risk used C-index, which asks whether the higher predicted risk precedes the earlier observed event among comparable pairs while accounting for censoring. Spearman and incremental-R^2 nulls were 0; AUROC and C-index nulls were 0.5. These measures assess association or discrimination, not probability calibration, causal mechanism, or clinical benefit; it is not a calibrated survival probability.

The saved discovery family used two-sided Spearman or Mann–Whitney tests and Benjamini–Hochberg false-discovery-rate control. Grade-adjusted audits used nested outer patient-disjoint evaluation and a grade-stratified permutation test of held-out increment. The full fitting specifications, nulls, multiplicity family, site resampling, and plain-language statistical definitions are retained in Supplementary Sections S2, S3, S6, and S7.

5.4 Conditional, setting, and outcome qualification

PTEN and AR conditional analyses compared clinical-only, image-only, and combined models in nested patient-disjoint folds. Primary held-out increments were change in AUROC for PTEN and change in R^2 for AR. Here R^2 is the proportion of held-out variation explained, and the registered delta was combined minus grade-only; a positive delta therefore favors adding the image-derived score. The untouched outer fold estimates that increment without reusing the patients that selected or fitted the probe. AR site effects used slide-level Spearman correlation with patients resampled as clusters. Site labels remained entangled with acquisition and case mix and were not interpreted causally.

Encoder, scale, tile-budget, and seed repeats reused patients and folds. Seed summaries and 390 seed-matched contrasts therefore describe correlated analytic sensitivity rather than independent replication. The complete grid, ranges, null crossings, and paired contrasts are reported in Supplementary Section S8.

The post-hoc recurrence score was fitted with a Cox model in LEOPARD and applied unchanged to TCGA-PRAD. Reconstructed recurrence and official PFI were evaluated separately. Paired clinical-versus-image comparisons reused the same complete-case patients and bootstrap draw; positive C-index or integrated Brier score improvement favored the augmented model under the registered delta definition. C-index improvement was augmented minus reference. Because a lower integrated Brier score indicates smaller time-dependent prediction error, its delta was defined in the reverse direction—reference minus augmented—so that positive consistently favored augmentation. The PCA sweep was exploratory and not an unbiased model-selection estimate. Endpoint definitions and all paired comparisons are provided in Supplementary Section S9.

5.5 Secondary ISUP functional-sensitivity extension

After the six targets had been evaluated at the representation level, the FM6 extension tested whether the most developed eligible coordinate affected an internal downstream head. It used 27,968 paired whole-tissue tiles aggregated to 392 TCGA-PRAD patients, including 80 BCR events. CONCH and Virchow were evaluated separately under the same five outer patient-disjoint folds. Within each training fold, 64 principal components summarized the representation, standardized ridge regression learned an ISUP direction, and a ridge-penalized Cox model learned BCR risk. Regularization was selected from the fixed grid 0.1, 1, 10, 100, and 1,000 using training data only.

For the primary intervention, the fitted Cox coefficients were locked while the normalized ISUP-probe direction was projected out of held-out representations. The estimand was full minus erased C-index. One hundred variance-matched random directions and 20 label-permuted directions served as controls; the one-sided empirical p value compared targeted erasure with the matched-random distribution. A secondary analysis refitted the Cox head after erasure to test compensation. Patient-bootstrap intervals used 2,000 resamples, and ISUP recoverability used 200 label permutations.

An ISUP-plus-AI versus ISUP-only comparison was a limited internal boundary check, not a comprehensive clinical-increment study. The analysis remained whole-tissue because the tumor detector's locked threshold failed the prespecified sensitivity gate in both newly held-out PANDA providers. Supplementary Section S5A reports the complete intervention and detector results and the associated claim ceiling. No equivalent head intervention was performed for the other five axes; their distinct eligibility and data limitations are reported in Table 1 and the Discussion.

For manuscript closure, the FM6 analysis was rerun under the locked protocol with the same 392 patients, 80 events, saved paired embeddings, five folds, random seed, preprocessing, regularization grids, resampling counts, controls, endpoints, and output schema. No tuning or data-dependent change was permitted. Before execution, the SHA-256 hashes of 20 protocol-defined nonvolatile outputs were retained as the baseline; volatile execution time and device metadata were excluded from this comparison. The analysis and figure-rendering entry points then regenerated the outputs, and handoff required exact equality of all 20 hashes plus unchanged headline estimates and gate states. A mismatch would have stopped manuscript handoff rather than triggered retrospective tuning. This procedure tests exact computational reproducibility of the internal analysis, not external biological replication.

The FM6 gate labels separate four questions: R, recoverability of ISUP from the frozen whole-tissue representation; A, internal association of the locked head with BCR; U, functional sensitivity of that head to targeted ISUP-direction removal; and T, equivalent external transport. R, A, and U passed at the stated internal exploratory level for both encoders, T was not tested, and the protocol therefore prohibited a strong H2 functional-use claim.

5.6 Uncertainty, missingness, and provenance

Where supported by the saved source, estimates used 95% percentile patient-bootstrap intervals; the patient bootstrap resamples patients rather than slides. AR site intervals used patient-cluster resampling. Unavailable intervals, event counts, single-class sites, undefined replicates, and legacy bootstrap accounting were preserved rather than reconstructed. An interval containing the registered null was treated as unresolved evidence, not proof that the biological relationship is absent; an interval excluding the null did not substitute for transport, functional-use, or clinical-utility evidence. The Benjamini–Hochberg procedure controlled the false-discovery rate across the saved discovery family, whereas target-specific bootstrap intervals described estimation uncertainty. No multiple imputation was performed for the paired recurrence analysis. Supplementary Section S10 provides the complete accounting.

Analyses used the repository's locked Python environment, fixed seeds, and auditable entry scripts. The builder verifies owner, byte count, and SHA-256 hash for every source registered in `provenance/source_evidence_manifest.csv`; figures and tables are regenerated from saved source tables, and a mismatch fails the build. Claim, endpoint, and numeric registries provide row-level lineage. The prostate-biomarker project remains owner of the frozen source analyses, while QFM owns this derivative manuscript.

Large language model assistance. OpenAI Codex assisted with code editing, test authoring, figure-code preparation, and language revision. The authors reviewed and verified all AI-assisted outputs and take full responsibility for the final manuscript.

5.7 Ethics approval and consent to participate

This work was a secondary analysis of deidentified data obtained from previously published public or access-governed repositories. No new participants were recruited, no intervention or specimen collection was performed, and the authors had no access to direct identifiers. The original studies obtained ethical approvals and informed consent or waivers as reported by the source publications and repositories. No additional ethical approval or participant consent was required for this secondary analysis. All data were used in accordance with the applicable repository access and data-use requirements. Consent for publication was not applicable because the manuscript reports aggregate results and contains no identifiable participant information.

Data Availability

The public cohorts used in this study retain their source-specific access and reuse terms. Digital Object Identifiers (DOIs) provide the persistent repository links below. NADT-Prostate is available from The Cancer Imaging Archive at `https://doi.org/10.7937/TCIA.JHQD-FR46` [11]; PANDA data are distributed through the PANDA challenge under its platform terms [10]; TCGA-PRAD histology and molecular data are available through the Genomic Data Commons, with the referenced TCGA clinical endpoint fields available through cBioPortal; and PRECISE is available from Zenodo at `https://doi.org/10.5281/zenodo.20721779` [17]. LEOPARD is an access-governed challenge resource: its training data and outcome labels remain subject to the LEOPARD challenge and dataset terms, and its public challenge report is cited here [15]. Publication-facing aggregate numerical source tables used to generate the article's figures and tables are supplied with the public code repository. Patient-level derived analysis artifacts and local provenance manifests are not redistributed.

Code Availability

The analysis-display code, aggregate source-data tables, provenance registries, and figure-reproduction builder are available at `https://github.com/AiXLab-GNU/evidence-qualified-alignment-prostate-cancer` under the annotated release tag `v1.0.2-submission`. Reference copies of the submitted main and Supplementary Information PDFs are included; editable manuscript source is maintained by the authors and is not publicly redistributed. The frozen source analyses remain maintained in `https://github.com/AiXLab-GNU/vlm-pathology`; their principal auditable entry points are

- `projects/prostate_biomarker_validation/code/legacy/build_revision_p0_artifacts.py`;
- `projects/prostate_biomarker_validation/code/legacy/aggregate_stability_grid.py`;
- `projects/prostate_biomarker_validation/code/legacy/build_tcga_cdr_pfi_evidence.py`;
- `projects/prostate_biomarker_validation/code/legacy/build_marker7_survival_paired_analysis.py`;
- `projects/prostate_biomarker_validation/code/legacy/run_marker7_common_source_sensitivity.py`; and
- `projects/prostate_biomarker_validation/code/legacy/build_ar_spop_evidence_closure.py`.

The auditable public-artifact entry point is `build_publication_artifacts.py`; its release tests are in `tests/test_publication_artifacts.py`. The builder verifies the recorded owner, byte count, and SHA-256 hash of every supplied aggregate source table before regenerating the publication figures and verifies the mapped headline values. No source model was retrained or refitted solely for the present interpretive re-analysis.

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Author Contributions

J.H.K. had overall responsibility for the manuscript, conceived and designed the study, conducted the experiments, and wrote the manuscript. D.H.S. contributed to the study concept and its verification, assessed the medical validity of the study, and reviewed the manuscript. H.C. reviewed the manuscript and curated the data. I.L. assessed the overall scientific integrity of the manuscript.

Additional Information

Competing Interests

The authors declare no competing interests.

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